Homework 23.10.2026 – Exercise 2.5

2.5 Let $\Omega = [0,1]$ and $\mathcal{F}_3$ = all countable subsets of Ω and all subsets of Ω having a countable complement.

(a) Show that $\mathcal{F}_3$ is a σ-algebra.

(b) Let us define $P(A) = 0$ if A is countable, and $P(A) = 1$ if A has a countable complement. Is $(\Omega, \mathcal{F}_3, P)$ a legitimate probability space?

$\Omega = [0,1]$

$\mathcal{F}_3 = \{A \subseteq \Omega;\ A \text{ is countable or } A^c \text{ is countable}\}$

(a) Is it a σ-algebra?

1. $\emptyset \in \mathcal{F}_3$ since it is countable.
2. $A \in \mathcal{F}_3 \Rightarrow A^c \in \mathcal{F}_3$: Let $B \in \mathcal{F} \Rightarrow$ either B or B^c is countable.
   if B is countable, then $B^c \in \mathcal{F}$, since $(B^c)^c = B$ is countable.
   if B^c is countable, then B^c is in $\mathcal{F}$ by definition.
3. $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$ $\forall A_i \in \mathcal{F}$: Let A_i, $i \in \mathbb{N}$ be elements of $\mathcal{F}$.
   if A_i is countable for $\forall i$, then $\bigcup_{i=1}^{\infty} A_i$ is also countable $\Rightarrow \bigcup A_i \in \mathcal{F}$
   if at least one of A_i, let's say A_j, is uncountable, its complement A_j^c must be countable and the union $\bigcup_{i=1}^{\infty} A_i$ is uncountable.
   $\left(\bigcup_{i=1}^{\infty} A_i\right)^c = \bigcap_{i=1}^{\infty} A_i^c$ (Morgan's law?) and A_j^c is countable $\Rightarrow \bigcap_{i=1}^{\infty} A_i = \left(\bigcup_{i=1}^{\infty} A_i\right)^c$ is countable $\Rightarrow \in \mathcal{F}$

(b) Yes.

$P(\Omega) = 1$ $\left(\Omega^c = \emptyset \text{ is countable}\right)$

Do we also check that P is a measure?

$P(\emptyset) = 0$, since $\emptyset$ is countable.

$P(A \cup B) = P(A) + P(B)$?

if both countable, $0+0=0$ ✓

if one has countable complement $0+1=1$ and $A \cup B$ has countable complement ✓

if both have countable complements, ?

↳ $A \neq B$ cannot both have countable complements, I think ...

(↳ Write rule more generally as $P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$ for A_i pairwise disjoint $A_i \in \mathcal{F}_3$

Let's suppose two of A_i, eg. B and C both have countable complements.

Then $P(B \cup C) = P(B) + P(C) = 1+1 = 2 > P(\Omega)$ which would mean P is not a measure

But can such $B, C \in \mathcal{F}_3$, $B \cap C = \emptyset$ exist?

B^c, C^c are countable $\Rightarrow B^c \cup C^c$ is countable

$(B^c \cup C^c)^c = B \cap C$ should then be uncountable and not $\emptyset$ ...

From $B \cap C = \emptyset \Rightarrow (B \cap C)^c = B^c \cup C^c = \Omega$ however cannot be true, since this should be countable →←